Latest in Digital Asset Cryptography -- diff report

Generated: 2026-08-31T17:40:05.456429 | **Period:** since 2026-08-31 | **Leaves with news:** 5 of 12

Fresh developments: 11 (as_of within 14 days of the window) | **Superseded/conflicting:** 4 | **Background captured:** 124 (older material, first time in the knowledge base)

Reconfirmations in this period (no new information, not shown): 2.

Summary

Leaf	Fresh	Superseded	Conflicting	Background
Exposure & migration analytics	3	0	2	19
Protocol proposals & activation politics	3	0	1	21
Protocol quantum roadmap	3	0	0	27
Institutional custody & exchanges	2	1	0	33
ECC attack demonstrations & resource estimates	0	0	0	24

Bitcoin

Exposure & migration analytics (`exposure-migration-analytics`)

New developments (3)

Dormant-coin monitoring

- In August 2026, six wallets untouched for more than ten years moved roughly $40 million, and most of that did not go to exchanges. ([coindesk.com — bitcoin wallets untouched for 10 years moved usd40 million most avoided exchanges](), as of 2026-08-28, confidence: medium, source: ingest)
- *New, dated dormant-coin movement event not previously recorded in this leaf.*
- The trend direction cuts against the quantum-panic narrative: dormant-coin activity is at its lowest level since 2022, and 2026 is on pace for under half of the previous year's total — old coins are moving less, not more. ([coindesk.com — bitcoin wallets untouched for 10 years moved usd40 million most avoided exchanges](), as of 2026-08-28, confidence: medium, source: ingest)
- *New trend data on dormant-coin movement frequency not previously recorded in this leaf.*

Exposed-pubkey UTXO estimates and methodology disputes

- Glassnode Research estimates 6.04 million BTC (30.2% of supply) quantum-exposed at rest as of August 2026, split as 1.92 million structurally exposed and 4.12 million operationally exposed. ([StarkWare's quantum-safe Bitcoin transaction reaches mainnet](), as of 2026-08-27, confidence: medium, source: grok)
- *New analytics shop (Glassnode) with a distinct lower total and structural/operational split not previously recorded.*

Conflicting -- needs review (2)

The estimates as a series, and where they diverge

- Coinbase's Independent Advisory Board on Quantum Computing and Blockchain, in a document dated 2026-06-08, states that "there are approximately 1.7 million Bitcoin spread across about 20,000 public keys that are of this form", referring to P2PK-style directly-exposed keys. The board includes professors from UT Austin, Stanford and UCSB alongside Ethereum Foundation and Coinbase representatives. (quantumzeitgeist.com — coinbase board details bitcoin keys, as of 2026-06-08, confidence: medium, source: ingest)
- conflicts with: "P2PK (about 1.7 million BTC, including Satoshi's estimated 1.1 million across roughly 22,000 addresses)"
- *Agrees on the 1.7M BTC total but gives a different address/key count (about 20,000 vs roughly 22,000), a methodology disagreement worth flagging rather than resolving.*
- CoinShares directly disputes higher figures: "Claims of 25% vulnerability often include temporary risks, such as reused exchange addresses, which can be easily mitigated." (coinshares.com — quantum vulnerability in bitcoin a manageable risk, as of 2026-02-06, confidence: medium, source: ingest)
- conflicts with: "The Risq List puts approximately 6.9 million BTC — roughly one third of circulating supply — in quantum-vulnerable addresses."
- *CoinShares explicitly challenges the methodology behind roughly-one-third-style vulnerability figures, a direct analytics-shop disagreement that should be flagged rather than resolved.*

Newly captured background -- older material, first time in the knowledge base (19)

Dormant-coin monitoring

- Satoshi Nakamoto's estimated 1.1 million BTC, worth over $85 billion, has been untouched since 2010, and prediction-market participants staked over $2.6 million on the question with odds strongly favouring continued dormancy through December 2026. (financefeeds.com — will satoshi nakamoto ever move his bitcoin, as of undated, confidence: low, source: ingest)
- *New dormant-coin monitoring detail (prediction-market activity and valuation) not previously recorded.*

Exposed-pubkey UTXO estimates and methodology disputes

- Project Eleven's Bitcoin Risq List reports 7,039,051 BTC at risk of quantum attack as of 2026-07-20 (block height 958,874). (Project Eleven - Bitcoin Risq List - FAQs, as of 2026-07-20, confidence: high, source: grok)
- *Newer, more precise data point from the same Risq List tracker (previously cited at ~6.9M as of 2026-05-06), extending the tracked series rather than merely restating it.*
- ChainQuery.com reports 7,052,314 BTC (~35.1% of circulating supply) at quantum risk as of 2026-08-02 (block 960,681), with 1,934,419 always-exposed and 5,117,895 reuse-exposed. (Bitcoin Quantum Exposure: 7.1M BTC at risk · ChainQuery.com, as of 2026-08-02, confidence: high, source: grok)
- *New analytics shop (ChainQuery) not previously represented in this leaf, with its own total and always-exposed/reuse-exposed breakdown.*

Taproot share

- Taproot's share of transactions peaked at approximately 42% in 2024, driven largely by Ordinals and inscriptions, then settled to roughly 15-20% of transactions by late 2025 as inscription activity declined. (spark.money — bitcoin segwit adoption tracker, as of undated, confidence: low, source: ingest)
- *First Taproot-adoption-share time series recorded in this leaf, distinct from the exposure-tier P2TR BTC figures already present.*

The canonical tracker and its breakdown

- Project Eleven maintains the Bitcoin Risq List, an open-source tracker of public-key exposure at the script and address-reuse level, and it is the closest thing to a canonical real-time public tracker of quantum-vulnerable BTC. (postquantum.com — project eleven quantum threat blockchains 2026, as of 2026-05-06, confidence:

medium, source: ingest)

- *Introduces the Risq List as a named, canonical tracker not previously recorded in this leaf.*
- The Risq List puts approximately 6.9 million BTC — roughly one third of circulating supply — in quantum-vulnerable addresses. (postquantum.com — project eleven quantum threat blockchains 2026, as of 2026-05-06, confidence: medium, source: ingest)
- *Adds a specific source attribution (Risq List) for the upper end of the baseline's undated 6.5-6.9M range, which was not yet attributed in this leaf.*
- That total decomposes as: address reuse about 4.99 million BTC (72.3% of the exposed total); legacy P2PK scripts about 1.72 million BTC (24.8%); Taproot P2TR about 198,000 BTC (2.9%), because the x-only public key is visible in the address; bare multisig negligible by comparison. (postquantum.com — project eleven quantum threat blockchains 2026, as of 2026-05-06, confidence: medium, source: ingest)
- *New granular breakdown of the exposed total by script category, not previously present.*
- Project Eleven's report was published 2026-05-06 and runs to 110 pages. (postquantum.com — project eleven quantum threat blockchains 2026, as of 2026-05-06, confidence: medium, source: ingest)
- *New provenance detail about the report underlying the Risq List figures.*

The estimates as a series, and where they diverge

- The same Coinbase board coverage separately attributes about 5 million BTC of address-reuse exposure to Project Eleven — consistent with the Risq List's 4.99 million, and confirming that the two organisations agree on the underlying counts. (quantumzeitgeist.com — coinbase board details bitcoin keys, as of 2026-06-08, confidence: medium, source: ingest)
- *Adds new corroborating detail that a second organisation (Coinbase's board) independently cites a figure matching Project Eleven's, which had not previously been recorded.*
- Galaxy Digital published a report on 2026-03-19 concluding Bitcoin's quantum vulnerability is significant but manageable, putting roughly 7 million BTC — about $470 billion at then-recent prices — at risk under a broad "long exposure" definition, and explicitly noting that other estimates come in lower depending on methodology. (coindesk.com — bitcoin s quantum threat is real but far from an existential crisis galaxy says, as of 2026-03-19, confidence: medium, source: ingest)
- *New estimate from a distinct analytics shop (Galaxy Digital) with its own methodology label, extending the data series.*
- CoinShares, in research published 2026-02-06, counts only about 1.6 million BTC in legacy P2PK addresses — about 8% of supply — as theoretically exposed, and narrows the practically significant figure much further: only 10,200 coins sit in UTXOs that "could cause any appreciable market disruption if they are stolen". (coinshares.com — quantum vulnerability in bitcoin a manageable risk, as of 2026-02-06, confidence: medium, source: ingest)
- *New, notably lower estimate and a distinct 'practically significant' narrowing methodology from a different analytics shop.*
- CoinShares' narrowing argument is a distributional one: the remaining ~1.6 million vulnerable coins are spread across roughly 32,607 individual UTXOs of about 50 BTC each, which they argue would take "millenia to unlock even in the most outlandishly optimistic scenarios". (coinshares.com — quantum vulnerability in bitcoin a manageable risk, as of 2026-02-06, confidence: medium, source: ingest)
- *New supporting methodological detail behind CoinShares' narrower risk figure.*
- In earlier research dated 2025-08-05 CoinShares set out the same methodology in more detail, finding that "almost all vulnerable coins are coinbase transactions that have never moved", that 34,068 P2PK addresses hold balances between 49 and 51 BTC, and that on their reading only about 24 addresses warrant serious concern. (coinshares.com — burning quantum vulnerable coins is a bad idea, as of 2025-08-05, confidence: medium, source: ingest)
- *Earlier data point in CoinShares' own series, with a distinct address-count granularity not previously recorded.*
- A widely-repeated 6.7 million BTC figure (about $600 billion) is given for coins vulnerable to a cryptographically relevant quantum computer through either unsafe UTXO type or address reuse, with up to 2.3 million BTC of that considered dormant and therefore an especially attractive target. (bitcoinsuisse.com — bitcoin takes first

computation and took hours to produce. (cointelegraph.com — starkware quantum resistant bitcoin transaction mainnet, as of 2026-08-27, confidence: medium, source: ingest)
- *New concrete milestone: the first real-world QSB deployment on mainnet.*
- That transaction was nonstandard and could not propagate through normal node relay — it had to be submitted directly through MARA Pool's Slipstream service. StarkWare characterised it as a last-resort measure rather than a replacement for protocol-level protection, and as applying to individual transactions rather than being a network-wide upgrade. (cointelegraph.com — starkware quantum resistant bitcoin transaction mainnet, as of 2026-08-27, confidence: medium, source: ingest)
- *New qualifying detail on how the QSB spend was propagated and its limited scope, contextualizing it against protocol-level proposals like BIP-360/361.*

Conflicting -- needs review (1)

BIP-361 — the legacy-signature sunset and its phases

- Press coverage dates the proposal's public submission to 2026-04-14, when Jameson Lopp and five co-researchers submitted the draft to the bitcoin/bips repository. This is a different date from the preamble's "Assigned: 2026-02-11" — see Contested. (theblock.co — bitcoin researchers propose phased sunset of legacy signatures to address quantum risks, as of 2026-04-14, confidence: medium, source: ingest)
- conflicts with: "BIP-361 (Post Quantum Migration and Legacy Signature Sunset) was published 2026-04-14"
- *The claim itself flags a discrepancy between the baseline's 2026-04-14 publication date and the repo preamble's 2026-02-11 assigned date, which needs human resolution rather than silent adoption.*

Newly captured background -- older material, first time in the knowledge base (21)

Activation politics and covenant interaction

- BIP-110 ("Reduced Data Temporary Softfork"), the live activation contest of August 2026, opened its mandatory signaling window at block 961,632 around 2026-08-09, running through block 963,647. It is unrelated to quantum but is the current working example of how a contested soft fork actually fares. (aminagroup.com — bitcoin fork august 2026 bip 110 ecash covenants and the quantum clock, as of 2026-08-09, confidence: medium, source: ingest)
- *New concrete example of contested soft-fork activation mechanics, relevant to activation politics.*
- As of 2026-07-30 BIP-110 signaling had reached only about 2% against a 55% early lock-in threshold, with no major mining pool committed and only Ocean producing meaningful signal; the expected outcome was a small breakaway chain rather than a network-wide change. (aminagroup.com — bitcoin fork august 2026 bip 110 ecash covenants and the quantum clock, as of 2026-07-30, confidence: medium, source: ingest)
- *New quantitative data on BIP-110's low signaling support and expected outcome.*
- OP_CAT (BIP-347) reached "Complete" specification status on 2026-03-01 but has no proposed mainnet activation parameters. OP_CTV (BIP-119) has a published activation client specifying a 90% miner threshold and a minimum activation height around May 2027. (aminagroup.com — bitcoin fork august 2026 bip 110 ecash covenants and the quantum clock, as of 2026-03-01, confidence: medium, source: ingest)
- *New status detail on both covenant proposals (OP_CAT, OP_CTV) not previously recorded.*
- The covenant proposals and the quantum migration are on independent timelines: neither covenants nor quantum-resistant spending paths have activation consensus, and their schedules are not aligned to each other or to any near-term fork. Quantum migration is characterised as a multi-year programme requiring the largest coordinated key rotation Bitcoin has ever attempted, and as not having properly started. (aminagroup.com — bitcoin fork august 2026 bip 110 ecash covenants and the quantum clock, as of undated, confidence: low, source: ingest)
- *New characterization of the relationship between covenant and quantum-migration timelines, and their lack of coordination.*

BIP-360 — status, naming, and what it actually specifies

- BIP-360's current title in the merged bitcoin/bips master copy is "Pay-to-Merkle-Root (P2MR)", its status is Draft, and its authors are Hunter Beast, Ethan Heilman and Isabel Foxen Duke. The preamble records it as assigned 2024-12-18. (github.com — bip 0360.mediawiki, as of undated, confidence: low, source: ingest)
- *New preamble-level detail (title, authors, assigned date) not previously recorded.*
- The merged BIP-360 text carries version 0.12.1 dated 2026-07-24 — the proposal has continued to be revised in-repository for over five months after the February merge, so "merged" should not be read as "frozen". (github.com — bip 0360.mediawiki, as of 2026-07-24, confidence: medium, source: ingest)
- *Adds new information about ongoing post-merge revision activity, beyond the baseline merge-date fact.*
- P2MR commits solely to the Merkle root of a script tree rather than to an internal public key, which removes Taproot's quantum-vulnerable key-path spend entirely; the draft specifies SegWit version 2, producing mainnet addresses beginning `bc1z`. (github.com — bip 0360.mediawiki, as of undated, confidence: low, source: ingest)
- *New technical detail on what P2MR specifies, not previously in knowledge base.*
- BIP-360 explicitly does NOT introduce post-quantum signatures. Its own text defers them, saying protection against more sophisticated quantum attacks "may require the introduction of post-quantum signatures in Bitcoin" and that this is worth considering "in the future"; it requires BIPs 340, 341 and 342 and mandates no PQ scheme. (github.com — bip 0360.mediawiki, as of undated, confidence: low, source: ingest)
- *New clarifying fact about scope of BIP-360, distinguishing it from a PQ signature proposal.*
- The output type has been renamed twice. A 2025-12-19 bitcoindev post by Hunter Beast announced a clean-sheet rewrite of BIP-360 — with Isabel Foxen Duke added as a third co-author to lead it — that renamed the output type from "Pay to Quantum Resistant Hash (P2QRH)" to "Pay-to-Tapscript-Hash (P2TSH)" for clarity, moved post-quantum signatures out to a future BIP, and added Python and Rust test vectors. The name in merged master is now P2MR, so the sequence is P2QRH → P2TSH → P2MR. (groups.google.com — 3JWQwnMUetg, as of 2025-12-19, confidence: medium, source: ingest)
- *New history of the naming evolution behind BIP-360's output type, not previously recorded.*
- P2MR supports multisig, timelocks, conditional payments, inheritance schemes and advanced custody; the cost is slightly larger transactions, since every spend is a script-path spend carrying more witness data than a Taproot key-path spend. (cointelegraph.com — bitcoin s quantum upgrade path what bip 360 changes and what it does not, as of undated, confidence: low, source: ingest)
- *New functional and cost detail about P2MR not previously captured.*

BIP-361 — the legacy-signature sunset and its phases

- BIP-361 "Post Quantum Migration and Legacy Signature Sunset" is in bitcoin/bips master with Layer "Consensus (soft fork)", Status Draft, Type Informational, and Assigned date 2026-02-11. Its authors are Jameson Lopp, Christian Papathanasiou, Ian Smith, Joe Ross, Steve Vaile and Pierre-Luc Dallaire-Demers, and its Requires field points at a still-unwritten "TBD Post Quantum Signature BIP". (github.com — bip 0361.mediawiki, as of undated, confidence: low, source: ingest)
- *New preamble-level detail on BIP-361 (layer, type, authors, requires field) not previously recorded.*
- Phase A forbids sending funds to quantum-vulnerable address types, permitting only legacy-to-post-quantum transfers, and activates approximately 160,000 blocks (about three years) after BIP-361 itself activates. (github.com — bip 0361.mediawiki, as of undated, confidence: low, source: ingest)
- *New specific mechanics of BIP-361's Phase A timeline.*
- Phase B tightens ECDSA/Schnorr signature validation at a predetermined block height roughly two years after Phase A, encumbering legacy signatures and requiring widespread wallet upgrades by that point. (github.com — bip 0361.mediawiki, as of undated, confidence: low, source: ingest)
- *New specific mechanics of BIP-361's Phase B timeline.*
- After Phase B, holders of ECDSA/Schnorr-secured UTXOs would spend only through "rescue protocols" built on asymmetric knowledge such as BIP-32 hardened key derivation, intended to let a genuine owner prove ownership while a quantum attacker holding only the public key cannot. (github.com — bip 0361.mediawiki, as of undated, confidence: low, source: ingest)

- *New detail on the post-Phase-B rescue-protocol mechanism, not previously recorded.*

Signature-scheme selection

- The scheme choice is a three-way trade with no decision taken, because BIP-360 defers signatures entirely: ML-DSA is FIPS-finalised (August 2024) and well-studied but large on-chain; FN-DSA-512 offers roughly a 3x reduction in on-chain footprint but its standard, FIPS 206, is not yet published; SLH-DSA rests on hash-function security rather than lattice assumptions and is the conservative backstop, at a larger signature size. (postquantum.com — fixing bitcoin pqc migration, as of undated, confidence: low, source: ingest)
- *New comparative detail on the three-way signature-scheme trade-off, going beyond the bare fact of FN-DSA's draft status.*

The freeze-or-let-them-burn debate

- Jameson Lopp's position is that freezing dormant coins is preferable to letting a future quantum attacker recover and dump them, and he characterises an attacker sweeping dormant coins as "theft from everyone". (theblock.co — bitcoin researchers propose phased sunset of legacy signatures to address quantum risks, as of undated, confidence: low, source: ingest)
- *New named position and quote in the freeze-or-burn debate.*
- Michael Saylor's stated position, posted on X on 2025-12-17, is that quantum computing will not break Bitcoin but "harden" it: "The network upgrades, active coins migrate, lost coins stay frozen. Security goes up. Supply comes down." (bitcoinist.com — saylor lost bitcoin freeze quantum risk, as of 2025-12-17, confidence: medium, source: ingest)
- *New named position and quote in the freeze-or-burn debate.*
- The objection to both positions is governance-shaped rather than technical: critics dispute who decides which coins are genuinely lost versus merely dormant, and argue that any freeze is inconsistent with self-custody and a fixed supply. (bitcoinist.com — saylor lost bitcoin freeze quantum risk, as of undated, confidence: low, source: ingest)
- *New framing of the counter-argument to freeze proposals, not previously captured.*
- Charles Hoskinson has argued publicly (2026-04-16) that Bitcoin's quantum fix amounts to a hard fork and cannot save Satoshi-era coins. Treat as a rival-ecosystem principal commenting on Bitcoin, not a neutral technical assessment. (coindesk.com — cardano s hoskinson says bitcoin s quantum fix is a hard fork that can t save satoshi s coins, as of 2026-04-16, confidence: medium, source: ingest)
- *New named outside commentary adding to the debate's public record.*

The no-soft-fork alternative (QSB) and its first mainnet spend

- On 2026-04-09 Avihu Levy of StarkWare published "QSB: Quantum Safe Bitcoin Transactions Without Softforks", describing a hash-based one-time signature scheme embedded in Bitcoin Script that works under existing consensus rules with no protocol change, no miner signaling and no activation timeline. (theblock.co — starkware quantum safe bitcoin, as of 2026-04-09, confidence: medium, source: ingest)
- *Introduces a genuinely new no-soft-fork alternative approach not previously recorded in the knowledge base.*
- QSB's cost is the point of the design, not a footnote: it requires substantial off-chain GPU computation, with estimated costs reported at roughly $75-$200 per transaction. (coindesk.com — quantum safe bitcoin now possible without a soft fork but costs usd200 a pop, as of 2026-04-10, confidence: medium, source: ingest)
- *New cost detail on the QSB proposal not previously recorded.*

Ethereum & L2s

Protocol quantum roadmap (`protocol-roadmap`)

New developments (3)

EIP-8141 / account abstraction for PQ

- EIP-8141 has a dedicated tracking site (eip8141.io) documenting the divergence between the original Jan 29, 2026 spec and the evolved current spec, and lays out a seven-stage PQ roadmap running from decoupling ECDSA to private L1 settlement. (Post-Quantum Roadmap | EIP-8141, as of 2026-08-31, confidence: high, source: grok)
- *New information about a dedicated tracking resource and a previously unrecorded seven-stage roadmap structure for EIP-8141's evolution.*

account-abstraction as the key-rotation escape hatch

- A new ethresear.ch proposal (Aug 28, 2026) suggests updating the ecrecover opcode to support post-quantum signatures from EIP-8141 frame transactions via sentinel values (v=27, s=0) and lookup encoding, without disabling the opcode. (Proposed PQ upgrade for ecrecover - Cryptography - Ethereum Research, as of 2026-08-28, confidence: high, source: grok)
- *Genuinely new research thread proposing a specific ecrecover-compatibility mechanism for PQ signatures, not previously recorded.*

planned forks with PQ preparation

- EIP-8141 is targeted for the Hegotá fork (H2 2026) as CFI (Considered for Inclusion) and was formally added to the EIP-8081 meta-EIP via PR #11537, merged April 30, 2026. (Post-Quantum Roadmap | EIP-8141, as of 2026-08-31, confidence: high, source: grok)
- *Adds new procedural detail (CFI status, meta-EIP inclusion via specific merged PR) beyond the prior general fork-assignment claim.*

Newly captured background -- older material, first time in the knowledge base (27)

Account abstraction as the key-rotation escape hatch

- EIP-8141 "Frame Transaction" is Standards Track (Core), Status Draft, created 2026-01-29, authored by Vitalik Buterin, lightclient, Felix Lange, Yoav Weiss, Alex Forshtat, Dror Tirosh, Shahaf Nacson, Derek Chiang, Toni Wahrstätter and Stavros Vlachakis. (eips.ethereum.org — eip 8141, as of 2026-01-29, confidence: medium, source: ingest)
- *First record of EIP-8141's formal metadata, status, and authorship.*
- EIP-8141 introduces a transaction type that abstracts validation, execution and gas payment through a "frame" system, decomposing transactions into contract calls that determine their own validity and payment mechanics rather than relying on fixed ECDSA keys. (eips.ethereum.org — eip 8141, as of 2026-01-29, confidence: medium, source: ingest)
- *New technical detail about how EIP-8141's frame mechanism decomposes transactions, not previously recorded beyond its metadata.*
- The EIP states explicitly that "it provides a native off-ramp from the elliptic curve based cryptographic system used to authenticate transactions today, to post-quantum (PQ) secure systems", and that unlinking accounts from specific elliptic-curve keys enables native key rotation. (eips.ethereum.org — eip 8141, as of 2026-01-29, confidence: medium, source: ingest)
- *New direct textual evidence of EIP-8141's stated PQ off-ramp rationale, not previously captured verbatim.*
- The EIP-8141 document itself does not name the fork that will carry it, even though the roadmap page assigns it to Hegotá. (eips.ethereum.org — eip 8141, as of 2026-01-29, confidence: medium, source: ingest)
- *New nuance clarifying that the fork assignment for EIP-8141 comes only from the roadmap page, not the EIP itself, adding precision without contradicting the earlier claim.*
- Ethereum's stated strategy is "signature agility" through account abstraction rather than a protocol-wide mandatory migration: individual accounts voluntarily adopt post-quantum signatures independently, a deliberate architectural difference from BIP-361's deadline-and-sunset approach on Bitcoin. (ethereum.org —

quantum resistance, as of undated, confidence: low, source: ingest)

- *New characterization of Ethereum's voluntary migration strategy contrasted explicitly with Bitcoin's BIP-361 approach.*

Active research threads

- Post-quantum attestation aggregation remains an open research area with live proposals on ethresear.ch, including separator-based participation commitments and a folding-based approach to PQ signature aggregation. (ethresear.ch — 23639, as of undated, confidence: low, source: ingest)
- *New record of ongoing ethresear.ch research threads on PQ aggregation not previously captured.*
- A 2026 IACR ePrint, "Transient Quantum Resistance, with Application to Ethereum Consensus", addresses the same consensus-layer problem from the academic side. (eprint.iacr.org — 1660, as of undated, confidence: low, source: ingest)
- *New academic-source pointer on the consensus-layer PQ aggregation problem, distinct from the ethresear.ch forum threads.*

Devnet progression

- The PQ devnet sequence is: pq-devnet-0 (October 2025, initial client specs with modified 3SF-mini consensus); pq-devnet-1 (December 2025, integrate leanSig signing and verification plus basic aggregation); pq-devnet-2 (January 2026, leanMultisig aggregation integrated with performance metrics); pq-devnet-3 (February 2026, aggregation separated from block production); pq-devnet-4 (March 2026, active, recursive PQ signature aggregation using leanVM); pq-devnet-5 (planned, block-level aggregation proofs replacing 3SF-mini with a "PQ heartbeat"). (leanroadmap.org, as of 2026-03-01, confidence: medium, source: ingest)
- *New detailed devnet timeline not previously recorded.*
- Weekly interop devnets run with more than ten client teams. (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *New operational fact about devnet cadence and participation scale.*

Fork milestones and timeline

- The structured fork milestones target completion of core post-quantum infrastructure by approximately 2029: Milestone I is a PQ key registry for validators; Milestone J is PQ signature verification precompiles; Milestone L is PQ consensus attestations via leanVM; Milestone M is full PQ aggregation and safe blob commitments. (ethereum.org — quantum resistance, as of undated, confidence: low, source: ingest)
- *New milestone framework with target date not previously recorded.*
- The Hegotá fork, scheduled for H2 2026, implements EIP-8141 — making it the nearest-term fork on this roadmap that carries post-quantum preparation. (ethereum.org — quantum resistance, as of undated, confidence: low, source: ingest)
- *New fork-specific timeline fact directly answering which fork carries PQ prep.*
- The Post-Quantum team's own assessment is that "L1 protocol upgrades could be completed by 2029, with full execution-layer migration taking additional years beyond that" — the roadmap does not claim Ethereum is post-quantum in 2029, only that the L1 protocol work is. (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *New nuance distinguishing L1 protocol completion from full ecosystem migration timeline.*
- pq.ethereum.org gives rough markers (I, *J*, L, *M*) rather than fixed dates for those milestones. (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *New clarification on the tentative nature of milestone dating.*
- Reported scope of the migration includes six quantum-resistant signature schemes and 13 EVM precompiles, with full activation planned before 2030. (coindesk.com — vitalik buterin unveils ethereum roadmap to counter quantum computing threat, as of 2026-02-26, confidence: medium, source: ingest)
- *New quantitative scope figures for the migration not previously captured.*

Named components and their status

- The component set is: leanXMSS (hash-based validator signatures), leanVM (minimal zkVM for SNARK-based aggregation), leanSig (Rust implementation of the signature scheme), leanSpec (an executable Python specification, in use by approximately ten client teams), and leanMultisig (implements the minimal zkVM for aggregation). (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *First full inventory of the named lean-Ethereum components.*
- As of the Lean Consensus roadmap's status table: leanSig is integrated, with verification running at 39% of its target performance; leanMultisig is in testing at about 97% of target efficiency on M4 Max systems; leanVM is still in development. (leanroadmap.org, as of undated, confidence: low, source: ingest)
- *New performance/status metrics for individual components.*

The four vulnerable systems

- Ethereum's official roadmap identifies four cryptographic systems needing replacement: consensus-layer BLS signatures (validator votes, relying on elliptic-curve pairings); KZG commitments (data availability for rollups, also pairing-based); ECDSA account signatures on secp256k1, whose public keys are exposed after a transaction is sent; and application-layer ZK-proofs built on pairing-based SNARKs. (ethereum.org — quantum resistance, as of undated, confidence: low, source: ingest)
- *New enumeration of the four vulnerable cryptographic systems central to the roadmap.*
- The planned replacements are, respectively: leanXMSS plus leanVM for BLS; STARK-based or lattice-based commitment schemes for KZG; post-quantum schemes reached via account abstraction and EIP-8141 for ECDSA; and natural ecosystem adoption of STARKs, which are already quantum-resistant, for ZK-proofs. (ethereum.org — quantum resistance, as of undated, confidence: low, source: ingest)
- *New mapping of replacement technologies to each vulnerable system.*

The post-BLS aggregation problem and its solution

- The hard part is not quantum-safe signing but quantum-safe AGGREGATION: BLS's property of combining hundreds of thousands of validator signatures into one has no obvious quantum-safe equivalent, and post-quantum signatures are far larger to begin with. (blog.lambdaclass.com — ethereum signature schemes explained ecdsa bls xmss and post quantum leansig with rust code examples, as of undated, confidence: low, source: ingest)
- *New framing of the core technical challenge in the aggregation problem.*
- The size gap is roughly 3,000 bytes for a hash-based validator signature against 96 bytes for BLS. leanVM, a minimal zkVM acting as a SNARK-based aggregation engine, is the answer to that gap, compressing by about 250x. (ethereum.org — quantum resistance, as of undated, confidence: low, source: ingest)
- *New quantitative detail on the signature size gap and leanVM's compression ratio.*
- leanXMSS is quantum-safe because it depends only on hash-function security, which quantum computers weaken (Grover) but do not break (Shor). (ethereum.org — quantum resistance, as of undated, confidence: low, source: ingest)
- *New explanation of why leanXMSS resists quantum attack.*

The roadmap and who owns it

- Ethereum's post-quantum work is led by the Post-Quantum team within the Ethereum Foundation's Protocol cluster, with a parallel Protocol Snarkification team under Alex Hicks focused on formal verification. (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *First record of the organizational structure behind Ethereum's PQ work.*
- The Post-Quantum Security team was established in January 2026, and the roadmap ("Strawmap") was published at strawmap.org following an Ethereum Foundation workshop that month. (coindesk.com — vitalik buterin unveils ethereum roadmap to counter quantum computing threat, as of 2026-01-01, confidence: medium, source: ingest)

- *New founding date and origin document for the roadmap, not previously recorded.*
- Vitalik Buterin publicly confirmed the roadmap and its timeline via X on 2026-02-26. (coindesk.com — vitalik buterin unveils ethereum roadmap to counter quantum computing threat, as of 2026-02-26, confidence: medium, source: ingest)
- *New fact about public confirmation event distinct from the team's establishment.*
- The work is organised across three protocol layers: the execution layer (account abstraction, explicitly "without a disruptive 'flag day'"), the consensus layer (replacing BLS signatures with post-quantum alternatives), and the data layer (securing data availability with post-quantum cryptography). (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *First record of the three-layer structural framing of the roadmap.*
- It unfolds in three phases: readiness and infrastructure, gradual adoption, and protocol-level consolidation. (pq.ethereum.org, as of undated, confidence: low, source: ingest)
- *New phase framework not previously recorded.*

post-BLS signature aggregation research

- A June 2026 ethresear.ch post explores the design space for a Post-Quantum Public Key Registry for Ethereum validators, proposing a phased rollout (registry fork first, signature switch later) using leanVM/pqSNARKs for aggregation. (Exploring the Design Space for a Post-Quantum Public Key Registry for Ethereum Validators - Cryptography - Ethereum Research, as of 2026-06-01, confidence: high, source: grok)
- *Adds a concrete phased-rollout design proposal for the PQ key registry milestone (I) not previously captured beyond the milestone's existence.*

Custody & wallets

Institutional custody & exchanges (`institutional-custody`)

New developments (2)

Diligence and questionnaire pressure

- The US Department of the Treasury launched a Quantum-Readiness Task Force on August 24, 2026, including a dedicated digital assets workstream to coordinate with crypto custodians and infrastructure providers on post-quantum migration. (Banks Can Finish Post-Quantum Migration and Still Be Exposed: Treasury Targets Vendors, as of 2026-08-26, confidence: high, source: grok)
- *New regulator-level initiative not previously recorded, directly relevant to counterparty/regulator questionnaire pressure.*

MPC custody providers

- A cross-regional pilot program launched in August 2026 tests quantum-resistant wallets and transfers using MPC with ML-DSA-65 signatures on a NEAR testnet, involving banks Bison Bank and DK Bank with regulators from Abu Dhabi, Bhutan, and Malta as observers. (Banks and regulators launch quantum safe crypto pilot, as of 2026-08-24, confidence: high, source: grok)
- *New, previously unrecorded cross-institutional MPC/PQ pilot involving named banks and regulator observers.*

Superseded (1)

Coinbase — the largest custodian's posture

- Coinbase plans to deliver an automated quantum-safe signing pipeline within a year of July 2026, using secure enclaves and threshold cryptography, while coordinating with Bitcoin core developers on post-quantum migration. (Coinbase Builds Post-Quantum Custody System, Funds Bitcoin's Crypto Upgrade, as of 2026-07-23, confidence: high, source: grok)

- supersedes: "[d0001:13] Coinbase is reported to be building a quantum-resistant version of the system protecting approximately 99.9% of its custodial assets, and to have begun upgrading its key management infrastructure to support post-quantum cryptography."
- *Provides a dated, higher-confidence, and more specific account (concrete timeline, named technologies, Bitcoin-core coordination) of the same Coinbase PQ custody upgrade effort previously reported only vaguely and at low confidence.*

Newly captured background -- older material, first time in the knowledge base (33)

BitGo — the furthest along

- BitGo and Silence Laboratories announced on 2026-05-26 the completion of what they describe as the first post-quantum MPC transaction simulation by a regulated custodian. (investors.bitgo.com — default, as of 2026-05-26, confidence: medium, source: ingest)
- *First mention of BitGo/Silence Labs PQ-MPC simulation milestone; knowledge base is empty.*
- The simulation used Silence Laboratories' PQ-MPC protocol built on ML-DSA (FIPS 204). It was a simulation, not production, and no specific blockchain was named. (investors.bitgo.com — default, as of 2026-05-26, confidence: medium, source: ingest)
- *New technical detail on the algorithm and scope of the simulation.*
- BitGo CEO Mike Belshe framed the move as a planning-horizon shift: "Quantum computing has moved from theoretical discussion to an infrastructure planning priority for the digital asset industry." (investors.bitgo.com — default, as of 2026-05-26, confidence: medium, source: ingest)
- *New executive framing/statement not previously recorded.*
- No production rollout date was given for the BitGo/Silence Labs PQ-MPC work. The companies stated only that they "currently plan to continue developing and testing the solution with select customers". (investors.bitgo.com — default, as of 2026-05-26, confidence: medium, source: ingest)
- *New fact establishing the absence of a rollout timeline for this specific initiative.*
- On 2026-07-09 BitGo launched four Bitcoin quantum-risk tools: a Quantum Risk Score, a Fix Exposed Addresses workflow, a UTXO selection method prioritising coins by address, and updated default address-type controls. (tftc.io — bitgo quantum risk controls bitcoin custody, as of 2026-07-09, confidence: medium, source: ingest)
- *New product-launch fact, distinct milestone from the earlier MPC simulation.*
- The migration workflow "guides clients through moving funds from elevated-risk addresses into new addresses with stronger key hygiene" — address-reuse remediation, the category dominating exposure numbers. (tftc.io — bitgo quantum risk controls bitcoin custody, as of 2026-07-09, confidence: medium, source: ingest)
- *New detail clarifying the mechanism and category of remediation covered.*
- BitGo's tools explicitly do NOT cover Taproot and P2PK addresses, which expose public keys from creation and fall outside these tools entirely; BitGo frames the tooling as bridging measures pending protocol-level upgrades (BIP-360/361), not permanent solutions. (tftc.io — bitgo quantum risk controls bitcoin custody, as of 2026-07-09, confidence: medium, source: ingest)
- *New scope-limitation fact about the tooling's coverage gaps.*
- No client adoption metrics were disclosed for BitGo's quantum-risk tools, and adoption is explicitly identified as "the variable that determines whether this work matters". (tftc.io — bitgo quantum risk controls bitcoin custody, as of 2026-07-09, confidence: medium, source: ingest)
- *New fact about disclosure/adoption status of the tools.*

Coinbase — the largest custodian's posture

- Coinbase's Independent Advisory Board on Quantum Computing and Blockchain published its report on 2026-06-13, with named participants including Yehuda Lindell, Dan Boneh, Scott Aaronson, Justin Drake,

Sreeram Kannan and Dahlia Malkhi. ([theblock.co — coinbase quantum report flags exchange cold wallets among millions of bitcoin exposed by address reuse](), as of 2026-06-13, confidence: medium, source: ingest)
- *New institutional development: a named advisory board and report not previously recorded.*
- The Coinbase advisory board report notes that of roughly 5 million BTC exposed through address reuse, "large amounts sitting in cold wallets of known exchanges" are part of that exposure; no specific exchanges are named. ([theblock.co — coinbase quantum report flags exchange cold wallets among millions of bitcoin exposed by address reuse](), as of 2026-06-13, confidence: medium, source: ingest)
- *New, differently-scoped estimate (address-reuse exposure specifically implicating exchange cold wallets) distinct from the general 6.5-6.9M exposed-public-key range.*
- The board distinguishes reuse-exposed 'active funds' from the 1.7 million BTC in legacy P2PK Satoshi-era addresses; active funds can be moved by owners today, unlike lost coins. ([theblock.co — coinbase quantum report flags exchange cold wallets among millions of bitcoin exposed by address reuse](), as of 2026-06-13, confidence: medium, source: ingest)
- *New categorical distinction and figure not previously recorded.*
- The Coinbase board's two recommendations are to begin technical migration work immediately without waiting for governance debate, and to communicate more clearly on timelines; it declines to endorse any specific solution. ([theblock.co — coinbase quantum report flags exchange cold wallets among millions of bitcoin exposed by address reuse](), as of 2026-06-13, confidence: medium, source: ingest)
- *New policy-recommendation content from the report.*
- Coinbase is reported to be building a quantum-resistant version of the system protecting approximately 99.9% of its custodial assets, and to have begun upgrading its key management infrastructure to support post-quantum cryptography. ([news.bitcoin.com — a quantum computer could one day break crypto security coinbase is preparing now](), as of undated, confidence: low, source: ingest)
- *New, low-confidence claim about internal Coinbase custody upgrade work.*
- Coinbase's documented cold-storage practice includes key-hygiene mechanics relevant to quantum exposure: one-time-use or reusable vault keys depending on whether the private key was fully reconstituted online, with remaining funds moved to a new single-use key if reconstitution occurs; Coinbase may also move assets between cold wallet addresses for security maintenance without client consent. ([sec.gov — tm2534140d8 ex10 4](), as of undated, confidence: low, source: ingest)
- *New documented custody-practice detail from an SEC filing, not previously recorded.*

Copper and the wider provider field

- The institutional digital-asset security platform market in 2026 is led by Fireblocks, BitGo, Anchorage Digital, DFNS, Copper, Ripple Custody, Ledger Enterprise, Fordefi, Safe and Hex Trust. ([ridgewayfs.com — digital asset security platforms](), as of undated, confidence: low, source: ingest)
- *New market-landscape fact establishing the competitive field for this leaf.*
- Copper provides MPC custody, settlement and prime brokerage, combining MPC with optical air-gapping; its ClearLoop product enables off-exchange settlement to reduce counterparty risk. ([ridgewayfs.com — digital asset security platforms](), as of undated, confidence: low, source: ingest)
- *New descriptive fact about Copper's product architecture.*
- Copper has no active custody licence in the US or UK and faced a July 2026 deadline for MiCA authorisation in the EU — a licensing constraint, not a quantum one, but material to counterparty assessment. ([ridgewayfs.com — digital asset security platforms](), as of undated, confidence: low, source: ingest)
- *New regulatory-status fact about Copper relevant to institutional due diligence.*
- Silence Laboratories, BitGo's PQ-MPC counterparty, has separately launched a quantum-safe vault product for crypto custody — the supply side of custodian PQ capability is beginning to productise independently of the custodians themselves. ([news.bitcoin.com — silence labs launches quantum safe vault to secure crypto custody](), as of undated, confidence: low, source: ingest)
- *New product-launch fact from Silence Labs, distinct from the earlier joint BitGo simulation milestone.*

Diligence and questionnaire pressure

- Proposed regulatory frameworks for digital asset custody reference NIST PQC algorithms — specifically naming CRYSTALS-Kyber — and CNSA 2.0 compatibility as security considerations, but this is descriptive rather than prescriptive: no named SEC rule or deadline for digital-asset custodians was found. (postquantum.com — us pqc regulatory framework 2026, as of 2026-02-01, confidence: medium, source: ingest)
- *New fact establishing the regulatory-reference landscape and clarifying it is non-binding.*
- Cyber-insurance underwriters in 2026 are increasingly incorporating quantum readiness into risk-assessment questionnaires at policy renewal, and organisations without a coherent PQC transition plan may face higher premiums — the clearest documented commercial pressure mechanism found in this pass. (postquantum.com — us pqc regulatory framework 2026, as of 2026-02-01, confidence: medium, source: ingest)
- *New mechanism of quantum-readiness pressure on custodians not previously captured.*
- The G7 Cyber Expert Group issued a coordinated financial-sector roadmap in January 2026 placing 2026-2027 as the awareness, strategy-development and initial-planning phase across G7 financial institutions, with cryptographic discovery and inventory in 2027-2028. (postquantum.com — financial services, as of 2026-01-01, confidence: medium, source: ingest)
- *New regulator-level timeline fact relevant to the questionnaire-pressure subtopic.*
- The Alternative Investment Management Association publishes a due diligence questionnaire for digital asset funds covering fund service providers including custody; no evidence was found that it currently contains a post-quantum section. (aima.org — press release aima publishes due diligence questionnaire for digital asset funds, as of undated, confidence: low, source: ingest)
- *New checkable fact about the current absence of PQ content in a specific industry DDQ template.*
- Singapore's Cyber Security Agency released a Quantum-Safe Migration Handbook and Quantum Readiness Index self-assessment questionnaire in July 2026 to help organizations prepare for post-quantum cryptography migration. (Quantum-Safe Migration Handbook and Quantum Readiness Index | Cyber Security Agency of Singapore, as of 2026-07-16, confidence: high, source: grok)
- *New national regulator questionnaire/handbook not previously recorded, adding to the documented set of PQ readiness pressure mechanisms.*

ETF custodians and concentration

- BlackRock appointed Coinbase as custodian for the iShares Bitcoin Trust, and added Anchorage as a second custodian in April 2025 as part of ongoing risk management given the trust's growing size. (finance.yahoo.com — blackrock bitcoin custodian besides coinbase 160000659, as of undated, confidence: low, source: ingest)
- *First mention of ETF custodian multi-sourcing arrangement for IBIT.*
- IBIT held 734,762 BTC worth about $48 billion as of 2026-07-15. (ledn.io — top spot bitcoin etfs, as of 2026-07-15, confidence: medium, source: ingest)
- *New quantitative fact establishing the scale of assets under a single custodial arrangement.*
- Coinbase serves as bitcoin custodian and prime execution agent for several competing exchange-traded bitcoin products as well as other digital-asset companies, concentrating custody for most spot ETFs in one provider. (outlookindia.com — blackrock coinbase and the systemic custodian question is bitcoin safe in the post etf market, as of undated, confidence: low, source: ingest)
- *New systemic-concentration fact not previously captured.*
- Morgan Stanley selected Coinbase and BNY as custodians for its planned bitcoin product, extending the same concentration pattern to a new large sponsor. (finance.yahoo.com — morgan stanley picks coinbase bny 185219374, as of undated, confidence: low, source: ingest)
- *New instance of the custody-concentration pattern extending to another sponsor.*
- BlackRock broadened IBIT's quantum-computing risk disclosure in a revised prospectus filed 2025-05-09, expanding what had been a brief mention into a detailed section on how advances in quantum systems could undermine digital-asset security, and noting that implementing defences would require broad consensus across a decentralized network. (dlnews.com — blackrock beefs up quantum compute threats bitcoin investors, as of 2025-05-09, confidence: medium, source: ingest)
- *New disclosure-content fact about ETF prospectus quantum-risk language, not previously recorded.*

state.

- No machine with even 1% of that capacity exists, and the paper is a resource estimate, not a timeline prediction. (postquantum.com — google quantum bitcoin ecdlp, as of 2026-03-30, confidence: medium, source: ingest)
- *Adds an explicit framing note about the paper's status (estimate vs. timeline) not present in baseline.*

The Oratomic / Caltech / Berkeley neutral-atom claim

- The paper "Shor's algorithm is possible with as few as 10,000 reconfigurable atomic qubits", from Oratomic, Caltech and UC Berkeley, was published 2026-03-31 — the day after the Google cryptocurrency paper. (postquantum.com — 10000 qubits shors, as of 2026-03-31, confidence: medium, source: ingest)
- *Baseline only notes the ~10K qubit claim generically; this adds the paper title, authorship, and precise publication date.*
- Its ECC-256 figures are architecture-dependent: space-efficient 9,700-11,000 physical qubits; balanced 12,000-13,300; time-efficient about 26,000. Runtime ranges from 10 to 264 days depending on parallelisation, assuming a 1-millisecond cycle time. (postquantum.com — 10000 qubits shors, as of 2026-03-31, confidence: medium, source: ingest)
- *New granular resource and runtime figures not previously captured.*
- The roughly 50x reduction against Google's 500,000-qubit estimate comes from using high-rate quantum LDPC codes at about 30% encoding efficiency versus about 4% for surface codes; their lifted-product code encodes 1,480 logical qubits in 5,278 physical qubits. (postquantum.com — 10000 qubits shors, as of 2026-03-31, confidence: medium, source: ingest)
- *New technical explanation of the comparative qubit-count reduction, not previously present.*
- The paper's own acknowledged limitations are substantial: the authors describe their code-surgery construction as "a theoretically consistent existence proof, which may not be optimal" and note verification became computationally prohibitive for the largest codes; the assumed 1-millisecond cycle time is undemonstrated, with current systems at several milliseconds per cycle on far fewer qubits; and individually addressed qubits at 10,000+ scale with maintained error rates lack experimental proof. (postquantum.com — 10000 qubits shors, as of 2026-03-31, confidence: medium, source: ingest)
- *New caveat detail substantiating why baseline treats this as a contested claim rather than fact.*
- Some of the codes underpinning the result were located via "LLM-assisted heuristic computer search", an unvalidated construction method that the community will need to reproduce independently. (postquantum.com — 10000 qubits shors, as of 2026-03-31, confidence: medium, source: ingest)
- *New methodological caveat about how the codes were found, not present in prior deltas.*
- The scaling gap is nonetheless smaller than for superconducting approaches: current neutral-atom systems control 6,100 atoms, against a requirement for the same count with fault-tolerant operations — roughly a 2x scaling challenge. (postquantum.com — 10000 qubits shors, as of 2026-03-31, confidence: medium, source: ingest)
- *New quantitative comparison of current hardware capability versus requirement, not previously recorded.*
- The claim carries a documented conflict of interest: all nine authors are shareholders in Oratomic and six are employed by the company, making the paper simultaneously a scientific result and a roadmap for the company's hardware approach. It had not passed peer review as of this research. (coindesk.com — quantum computers could break crypto wallet encryption with just 10 000 qubits researchers say, as of undated, confidence: low, source: ingest)
- *New disclosure about author conflicts of interest and peer-review status, not previously captured.*
- The authors state that a neutral-atom machine with about 26,000 qubits could break ECC-256 in about 10 days. (coindesk.com — quantum computers could break crypto wallet encryption with just 10 000 qubits researchers say, as of 2026-03-31, confidence: medium, source: ingest)
- *Pairs the specific 26,000-qubit time-efficient variant with its fastest 10-day runtime, a correlation not explicitly stated in the prior general range figure [d0001:18].*

The Q-Day Prize demonstration

- Project Eleven awarded its Q-Day Prize — a one-bitcoin bounty worth approximately $78,000 at the time — to Giancarlo Lelli on 2026-04-24 for breaking a 15-bit elliptic curve key on publicly accessible quantum hardware. (thequantuminsider.com — project eleven q day prize quantum ecc attack, as of 2026-04-24, confidence: medium, source: ingest)
- *Baseline only records the award and 15-bit break; this adds the winner's identity and prize value, which is new detail.*
- Lelli derived a private key from its public key across a search space of 32,767 using a variant of Shor's algorithm, on a publicly accessible IBM quantum computer — no national lab and no private chip. (thequantuminsider.com — project eleven q day prize quantum ecc attack, as of 2026-04-24, confidence: medium, source: ingest)
- *New mechanism and hardware detail not present in baseline.*
- The result was presented as a 512x jump over the previous public demonstration and as the largest public demonstration to date of the attack class threatening Bitcoin, Ethereum and over $2.5 trillion in ECC-secured digital assets. (blog.projecteleven.com — project eleven awards 1 btc q day prize for largest quantum attack on elliptic curve cryptography to date, as of 2026-04-24, confidence: medium, source: ingest)
- *New quantitative framing (512x jump, $2.5T exposure) beyond baseline's bare mention of the award.*
- The demonstration was rebutted on the record: Jonas Schnelli, a former Bitcoin Core maintainer, reproduced the key recovery in Python using purely random bits, finding the quantum computer's outputs statistically indistinguishable from random coin flips. (news.bitcoin.com — ibm quantum hardware cracks 15 bit ecc key but bitcoin devs say random bits match the result, as of undated, confidence: low, source: ingest)
- *New skeptical-response information not covered anywhere in baseline or prior deltas.*
- Independent researcher Yuval Adam confirmed this by swapping the IBM quantum backend for Linux's /dev/urandom and obtaining identical results. Coinkite founder Rodolfo Novak argued that "the private key is classically solved before the quantum circuit even runs" and that the system relies on "a classical verify filter". Jimmy Song also criticised the result. (news.bitcoin.com — ibm quantum hardware cracks 15 bit ecc key but bitcoin devs say random bits match the result, as of undated, confidence: low, source: ingest)
- *New corroborating criticism detail, not previously recorded.*
- The mechanism of the objection is that a 15-bit search space contains only 32,767 candidate keys — small enough that classical verification against the public key finds a match by near-random sampling with high probability, so the quantum stage contributes no advantage. (news.bitcoin.com — ibm quantum hardware cracks 15 bit ecc key but bitcoin devs say random bits match the result, as of undated, confidence: low, source: ingest)
- *New technical explanation of the rebuttal, not previously recorded.*
- Project Eleven CEO Alex Pruden acknowledged the criticism, conceded the result was "not Q-Day" and that NISQ experiments depend on classical assistance, and reframed it as incremental progress on accessible hardware while maintaining that post-quantum migration planning remains prudent. (news.bitcoin.com — ibm quantum hardware cracks 15 bit ecc key but bitcoin devs say random bits match the result, as of undated, confidence: low, source: ingest)
- *New detail closing out the dispute narrative, not previously recorded.*

The exposure-window arithmetic

- The widely-quoted "9 minutes to crack Bitcoin" figure comes from the Google paper's circuit execution time: assuming microsecond-scale gate operations and accounting for execution overheads, the full circuit runs on the order of tens of minutes, but because portions can be precomputed before a specific key is targeted, the target-specific phase is estimated at approximately 9 minutes. (safeheron.com — google quantum whitepaper, as of undated, confidence: low, source: ingest)
- *New exposure-window math not present in baseline.*
- Nine minutes matters because it sits just inside Bitcoin's roughly 10-minute average block time, which would in principle allow an attacker to break a key from a public key revealed in the mempool before the transaction confirms. This is the short-range or "on-spend" attack, one modality among several rather than the general case. (safeheron.com — google quantum whitepaper, as of undated, confidence: low, source: ingest)